

National Measurement System 1997-2000 Valid Analytical Measurement (VAM) Programme

Report on priorities for
reference materials and
calibration standards for out-
of-laboratory measurements
made using portable
measuring equipment

April 1998

LGC/VAM/1998/019



*Setting standards
in analytical science*

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VALID ANALYTICAL MEASUREMENT

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1. Executive Summary

This report has been produced as a formal deliverable under the DTI's National Measurement System Valid Analytical Measurement (VAM) Programme. It describes the study undertaken by LGC to identify priorities for reference materials and calibration standards for out-of-laboratory environmental measurements made using portable measuring equipment.

Previous VAM surveys and studies, along with some of the main regulation and guidelines relating to environmental measurements have been reviewed to identify reference material requirements. Updated information on requirements has been gained through consultation with interested parties and, in particular, the suppliers and users of test kits and portable monitoring instruments. The most popular matrices and analytes, along with more detailed information on specific requirements have been identified.

It is clear that analytical measurements made using portable measuring equipment are not carried out exclusively in the field with samples often being submitted for laboratory analysis using traditional techniques and instrumentation. Although calibration standards appear to be used fairly widely, the current uptake and use of matrix reference materials is low, even though many out-of-laboratory measurements are undertaken in support of legislation, regulation and national guidelines. The impact of regulation on the need for reference materials has been reinforced during discussions with users who themselves have specified a requirement for materials which are representative of analyses undertaken in support of compliance.

Views on the benefits and requirements for reference materials are varied and range from those who do not consider reference materials to be of value to others who believe they are of benefit and have firm views on specific requirements. It is clear that there is a need to increase awareness of the role and application of reference materials in achieving valid analytical measurements.

2. Introduction

This report describes work carried out by LGC to identify and assess priorities for reference materials and calibration standards for out-of-laboratory measurements made using test kits and portable monitoring instruments.

The report has been produced as a formal deliverable under the DTI's National Measurement System Valid Analytical Measurement (VAM) Programme and, more specifically, as part of the LGC VAM Project on 'Out-of-Laboratory Measurement'. It constitutes LGC's delivery of the VAM Project 3.3.1 milestone reference 3.3.1b.

This introductory section summarises the background to the work along with the scope and strategy adopted during the study. The following two sections describe the background material reviewed and the output of the consultation exercise. This is followed by an analysis and discussion of the findings and the final section presents the conclusions and recommendations of the study.

2.1 Background

An increasing number of analytical measurements are being performed outside permanent laboratories using portable measuring equipment.

The validity of data derived from such measurements and equipment is clearly of vital importance, for example, to demonstrate compliance with environmental legislation. However, it is particularly difficult to obtain valid and reliable measurements outside the laboratory. The inability to control the environment in which the measurements are made and the use of untrained operators both have potential to impact significantly on the reliability of data. The situation is made worse because of the lack of adequate quality assurance and control procedures, the shortage of reference materials and calibration standards, and the lack of adequate specifications and performance data for the procedures and equipment employed.

The value of reference materials as a tool to improve the reliability and comparability of analytical measurement is well recognised. However, previous studies [1,2] have indicated that there is not only a lack of reference materials available for out-of-laboratory applications, but also that the uptake and use of materials for such measurements is lower than for laboratory-based analysis.

As part of the 1997-2000 VAM Project on 'Out-of-Laboratory Measurement', LGC has therefore carried out a study to identify priorities for reference materials and calibration standards for out-of-laboratory measurements and, in particular, those measurements made with portable measuring equipment. This report constitutes the formal output of the study.

2.2 Scope

Although out-of-laboratory measurements are undertaken across a broad range of industrial and analytical sectors and for a variety of reasons (e.g. in clinical, food and medical diagnostics; to control production process; and to monitor emissions and discharges to the environment), it was recognised at an early stage that the project would need to have a specific focus in order to be most effective.

Following a series of internal discussions within LGC, we reported [3] that 'environmental monitoring' had emerged as the preferred focus for the project. Subsequent meetings held with Robert Bogue (the consultant who undertook the VAM Out-of-Laboratory Formulation Study [1]) and Hugh Berridge (the VAM Working Group 'tracker' for this project) endorsed the decision to focus on this area.

Particular reasons for concentrating on the environmental monitoring area include:

- it is relatively immature but developing rapidly;

- it is driven by legislation and many measurements are made to ensure compliance with legislation;
- many monitoring practices are less well validated than in other more mature areas (e.g. occupational health and safety monitoring, medical diagnostics);
- new measurement technology is exerting a growing impact; and
- out-of-laboratory measurements are widespread.

Whilst air and gas monitoring is an integral part of the UK's overall environmental monitoring activity, it was felt that the predominant interest in this area rested with NPL and AEA Technology and their respective VAM projects on gas analysis and particulates. This study has, therefore, focused on priorities for reference materials in environmental monitoring of land and water.

Out-of-laboratory measurements are performed using a wide range of equipment (e.g. test kits, hand-held instruments, transportable instruments, fixed instruments and process analysers). The pre-defined focus for this project was on those measurements made using portable measuring equipment (i.e. test kits and hand-held instruments), rather than fixed instruments or process analysers.

2.3 Strategy / methodology

The overall aim of the study was to identify requirements and priorities for reference materials and calibration standards for out-of-laboratory measurement applications.

In order to meet this aim, the following specific objectives were identified:

- to review existing material and information relating to reference material needs; and
- to undertake a wider consultation to provide updated information on specific needs.

There is a wide range of existing material and information relating to the requirements and priorities for reference materials. Earlier studies, such as the survey carried out in the 1994-1997 VAM Programme, the study on the use of on-site analytical measurement, along with specific enquiries and requests made to LGC's Office of Reference Materials (ORM) and the Reference Materials Advisory Service (REMAS) provide a valuable source of information on specific needs. In addition, because many environmental measurements are undertaken in support of regulatory compliance, we considered that the legislation itself could provide a useful indication of the scope of analyses undertaken and the range of reference materials required. However, the needs of the analytical community are not static and requirements for reference materials change. We have, therefore, also undertaken a wider consultation, particularly amongst the suppliers and users of portable instruments, to seek updated information on specific needs.

In undertaking the study we also felt it was important to identify the purpose for which reference materials would be used. This was necessary because the type of reference material produced and, in particular, the level and extent of characterisation and certification required is often directly related to the purpose for which the material will be used. For example, materials which are used for validation and calibration purposes might be expected to require a higher degree of certification than those which are used as check samples or for routine analytical quality control.

Following agreement by the NMSPU, we engaged a consultant (Robert Bogue) to help deliver the study and supplement LGC's own in-house expertise in the environmental monitoring area. Where possible, we have also attempted to identify examples of analytical problems experienced in the field, as these too provide an indication of the type and properties of reference materials required. Our discussions with the suppliers and users of portable equipment have also identified a number of issues relating to the uptake of reference materials for use in the field. Although not initially within the planned scope of the study, we have taken the opportunity to review these and highlighted them in the report.

2.4 Abbreviations

The following abbreviations are used in this report:

OOL	Out-of-laboratory
EA	Environment Agency (UK)
USEPA	Environmental Protection Agency (US)
ORM	Office of Reference Materials (at LGC)
REMAS	Reference Materials Advisory Service (at LGC)
ICRCL	Interdepartmental Committee on the Redevelopment of Contaminated Land (UK)
NIST	National Institute of Standards and Technology (US)
SWG	Sensors for Water Interest Group (UK)
RM	Reference material
CRM	Certified reference material
BOD	Biochemical oxygen demand (of a water sample)
DO	Dissolved oxygen (in water)
VOCs	Volatile organic compounds
PAHs	Polycyclic aromatic hydrocarbons
PCBs	Polychlorinated biphenyls
THM	Trihalomethanes
IA	Immunoassay

3. Background information

3.1 Regulatory and legislative requirements

A high proportion of analytical measurements performed in the environmental monitoring sector are undertaken in support of international agreements, European Directives and United Kingdom legislation and regulations. As such, the legislation itself often provides a good indication of the common analyses performed and, therefore, of the types and properties of reference materials required.

This section highlights some key UK regulations and legislation which potentially impact on the need for reference materials to support environment monitoring and, in particular, for the analysis of land and water. It is not intended to be an exhaustive listing or review of all legislation in this area, but more as a means of illustrating the range of legislation and the variety of analytical measurements required to underpin it.

3.1.1 Land

Although DOE is believed to be considering the preparation of a priority list of pollutants in land, there is, unlike water, no current national legislation setting out prescribed limits or concentrations of pollutants or contaminants in land. At a generic level, site-specific risk-based assessments are the preferred means for establishing the requirement for investigation and remediation.

However, basic guidelines do exist. A draft British Standard [4] provides recommended procedures for on-site investigation and includes minimum analyses (see Table 1) which should be undertaken depending on the historical use of the site and potential hazards that might arise from the proposed end-use. Similarly, the Interdepartmental Committee on the Redevelopment of Contaminated Land (ICRCL) has produced guidance notes [5] which provide criteria for assessing hazards of sites being redeveloped against the proposed end-use through the use of ‘threshold’ and ‘action’ trigger values for specific contaminants (see Table 2). The Environmental Protection (Prescribed Processes and Substances) Regulations 1991 [6] also controls the discharge of dangerous substances to land (see Table 3).

Whilst these examples are by no means exhaustive, they provide a useful indication of the variety of analyses performed and, therefore, on the potential range of reference materials required for the environmental analysis of land.

Table 1 - Examples of minimum analyses identified in the draft British Standard (DD175:1998) Code of practice for the identification of potentially contaminated land and its investigation

Analyses		
• total arsenic	• sulphate	• mineral oils
• total cadmium	• sulphide	• ammonium ion
• total lead	• chloride	• methane
• total copper	• free cyanide	• asbestos
• total nickel	• PAHs	• radioactive materials
• total zinc	• phenols	• combustable materials (e.g. coal dust, oil, tar, pitch)
• sulphur	• oily and tarry substances	
<i>(Note: although these are indicative of the range of analyses which might be carried out, the nature and extent of specific analyses undertaken depends on the previous use of the land and potential hazards that may arise as a result of its proposed end-use.)</i>		

Table 2 - Examples of Interdepartmental Committee on the Redevelopment of Contaminated Land (ICRCL) threshold and action trigger values for contaminants

Contaminant	Threshold concentrations (mg/kg air dried soil)		Action concentrations (mg/kg air dried soil)	
arsenic	10	40	not specified	
cadmium	3	15	not specified	
chromium (vi)	25		not specified	
chromium (total)	600	1000	not specified	
lead	500	2000	not specified	
mercury	1	20	not specified	
selenium	3	6	not specified	
boron	3		not specified	
copper	130		not specified	
nickel	70		not specified	
zinc	300		not specified	
PAHs	50	1000	500	10000
phenols	5	5	200	1000
cyanide (free)	25	100	500	500
cyanide (complexes)	250	250	1000	5000
thiocyanate	50			
sulphate	2000	2000	10000	50000
sulphide	250		1000	
sulphur	5000		20000	
acidity	pH5		pH3	
(Notes: This table is intended to illustrate the potential range of analyses that might be undertaken, although the actual analyses performed and the relevance of specific trigger values will depend on the proposed end-use of the land.)				

Table 3 - The Environmental Protection (Prescribed Processes and Substances) Regulations 1991: Prescribed Substances for Release into Land

- organic solvents - halogens and their covalent compounds - organo-metallic compounds - polychlorinated dibenzofuran and any congener thereof - polyhalogenated biphenyls, terphenyls and naphthalenes - phosphorus	- azides - metal carbonyls - oxidising agents - polychlorinated dibenzo-p-dioxin and any congener thereof - alkali metals and their oxides and alkaline earth metals and their oxides - pesticides
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3.1.2 Water

The UK is committed under European Community law, by various international conventions, and through its participation in Ministerial conferences on the North Sea to control and monitor the discharge of a range of dangerous substances into the aquatic environment.

There are a host of international agreements, EC Directives, UK regulations and legislation which rely on reliable and comparable analytical measurements to underpin the protection of the aquatic environment and to ensure the quality of water we drink.

For example, the EC Directive relating to the quality of water intended for human consumption (80/778/EEC) [7] (implemented through The Water Supply (Water Quality) Regulations 1989) lists a range of parameters for which prescribed concentrations in drinking water must not be exceeded (see Table 4).

Table 4 - The Water Supply (Water Quality) Regulations 1989 - Parameters for which prescribed limits must not be exceeded

Parameter	Maximum Concentration (unless otherwise stated)
colour	20 mg/l Pt/Co scale
turbidity	4 Formazin turbidity units
odour	3 at 25°C (Dilution number)
taste	3 at 25°C (Dilution number)
temperature	25°C
hydrogen ion concentration	5.5<pH>9.5
conductivity	1500 $\mu\text{S cm}^{-1}$ at 20°C
chloride	400 mg/l Cl
calcium	250 mg/l Ca
total hardness	60 mg/l Ca (minimum)
alkalinity	30 mg/l HCO_3 (minimum)
sulphate	250 mg/l SO_4
magnesium	50 mg/l Mg
sodium	150 mg/l Na
potassium	12 mg/l K
dry residues	1500 mg/l (after drying at 180°C)
nitrate	50 mg/l NO_3
nitrite	0.1mg/l NO_2
ammonium	0.5 mg/l NH_4
kjeldahl nitrogen	1 mg/l N
oxidizability	5 mg/l O_2
TOC	no significant increase in C mg/l
dissolved hydrocarbons	10 $\mu\text{g/l}$
phenols	0.5 $\text{C}_6\text{H}_5\text{OH}$ $\mu\text{g/l}$
surfactants	200 $\mu\text{g/l}$ (as lauryl sulphate)

Parameter	Maximum Concentration (unless otherwise stated)
aluminium	200 µg/l Al
iron	200 µg/l Fe
manganese	50 µg/l Mn
copper	3000 µg/l Cu
zinc	5000 µg/l Zn
phosphorus	2200 µg/l P
fluoride	1500 µg/l F
silver	10 µg/l Ag
arsenic	50 µg/l As
cadmium	5 µg/l Cd
cyanide	50 µg/l CN
chromium	50 µg/l Cr
mercury	1 µg/l Hg
nickel	50 µg/l Ni
lead	50 µg/l Pb
antimony	10 µg/l Sb
selenium	10 µg/l Se
boron	2000 µg/l B
barium	1000 µg/l Ba
pesticides	0.1 µg/l (individual), 0.5 µg/l (total)
PAHs	0.2 µg/l
benzo 3,4 pyrene	10 ng/l
tetrachloromethane	3 µg/l
trichloroethene	30 µg/l
tetrachloroethene	10 µg/l
substances extractable in chloroform	1 mg/l dry residue

In addition, there is a range of other legislation and regulations which control the discharge of dangerous substances to the aquatic environment, for example:

- The Urban Waste Water Treatment (England and Wales) Regulations 1993 (see Table 5);
- The Environmental Protection (Prescribed Processes and Substances) Regulations 1991 (see Table 6); and
- The Surface Waters (Dangerous Substances)(Classification) Regulations 1997 [8] (see Table 7).

Whilst these examples are by no means exhaustive, they serve to illustrate the range of existing legislation which depends on valid analytical measurement and, therefore, of the potential diverse range of matrices and analytes for which reference materials are required.

Table 5 - Some parameters listed in the Urban Waste Water Treatment (England and Wales) Regulations 1993

Parameter	Concentration
biochemical oxygen demand (BOD)	25 mg/l O ₂
chemical oxygen demand (COD)	125 mg/l O ₂
total phosphorus	1-2 mg/l P
total nitrogen	10-15 mg/l N

Table 6 - The Environmental Protection (Prescribed Processes and Substances) Regulations 1991: Prescribed Substances for Release into Water

- mercury and its compounds - all isomers of hexachlorocyclohexane - pentachlorophenol and its compounds - hexachlorobutadiene - dieldrin - polychlorinated biphenyls - dichloroethane - atrazine - tributyltin compounds - trifluralin - azinphos-methyl - endosulfan	- cadmium and its compounds - all isomers of DDT - hexachlorobenzene - aldrin - endrin - dichlorvos - all isomers of trichlorobenzene - simazine - triphenyltin compounds - fenitrothion - malathion
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Table 7 - The Surface Waters (Dangerous Substances)(Classification) Regulations 1997: Prescribed Limits for Dangerous Substances

Dangerous Substance	Limit (inland freshwaters) (mg/l)	Limit (coastal waters) (mg/l)
arsenic	50	25
atrazine and simazine	2	2
azinphos-methyl	0.01	0.01
dichlorvos	0.001	0.04, 0.6*
endosulphan	0.003	0.003
fenitrothion	0.01	0.01
malathion	0.01	0.02
trifluralin	0.1	0.1
tributyltin	0.02*	0.002*
triphenyltin (& derivatives)	0.02*	0.008*

(Note: the above limits are for annual mean concentrations)

except those marked which are for absolute maximum concentrations)*

However, regulation and legislation does not stand still. New legislation, changes to the range of parameters listed, along with revisions to their prescribed or specified limits, impact not only on the range of analyses required, but also on the type and properties of reference materials needed to ensure reliable and comparable measurement.

Such changes are illustrated by the proposed revision to the EC Drinking Water Directive [9,10] which affects not only the number of parameters listed, but also the maximum permitted concentrations of several substances. In the new Directive it is proposed to reduce the total number of parameters listed from 67 to 48, although 13 new parameters have been added. The parameters listed in the Directive have also been split into three groups: microbiological parameters; chemical parameters; and indicator parameters. Full listings of the proposed indicator and chemical parameters along with their parametric values are presented in Table 8 and Table 9 respectively. Some of the main changes to parameters and the parametric values proposed in the new Drinking Water Directive are listed in Table 10.

Table 8 - Indicator parameters listed in the proposed new EC Drinking Water Directive

Parameter	Parametric value
aluminium	200 µg/l
ammonium	0.5 mg/l
colour	acceptable to consumers and no abnormal change
conductivity	2500 µS cm^{-1} at 20°C
dissolved oxygen	≈50 % saturation
hydrogen ion concentration	= 6.5 and = 9.5 pH units
iron	200 µg/l
manganese	50 µg/l
odour	acceptable to consumers and no abnormal change
oxidizability	5 mg/l O ₂
sulphate	250 mg/l
taste	acceptable to consumers and no abnormal change
total bacteria count	No abnormal change
total coliforms	0 (number/100ml)
total organic carbon (TOC)	4 mg/l C and no abnormal change
turbidity	acceptable to consumers and no abnormal change

The new Drinking Water Directive also impacts on the way in which monitoring will be carried out, with two types of monitoring, check and audit monitoring, being proposed. The purpose of check monitoring is to provide the information necessary to decide whether or not a water supply appears to be in conformity with the Directive's parametric values. The parameters measured are general indicators of water quality and analyses can either be carried out on-site or by simple laboratory procedures. The purpose of audit monitoring is to provide the information necessary to decide whether all of the Directive's parametric values are being respected. Because check and audit monitoring will be undertaken at different frequencies and will involve determination of different sets of parameters, it is possible that different types of reference materials might be required for the two types of monitoring.

Furthermore, a new approach to the use of reference methods of analysis has been proposed, permitting the use of any method meeting certain performance standards rather than requiring specific methods to be used. The new Directive requires that methods of analysis must be capable of measuring concentrations equal to the parametric values within specified limits of accuracy, precision and limit of detection. Clearly, this impacts on the requirement for certified reference materials as a means for validating procedures and ensuring that methods are within their required performance characteristics.

Table 9 - Chemical parameters listed in the proposed new EC Drinking Water Directive

Parameter	Parametric value
acrylamide	0.25 µg/l
antimony	3 µg/l
arsenic	10 µg/l
benzene	1 µg/l
boron	1 mg/l
bromate	10 µg/l
cadmium	5 µg/l
chromium	50 µg/l
copper	2 mg/l
cyanide	50 µg/l
1,2 -dichloroethane	3 µg/l
epichlorhydrin	0.5 µg/l
fluoride	1.5 mg/l
lead	10 µg/l
mercury	1 µg/l
nickel	20 µg/l
nitrate	50 mg/l
nitrite	0.1 mg/l
pesticides	0.1 µg/l (applies to individual pesticides)
pesticides (total)	0.5 µg/l
PAHs	0.2 µg/l (sum of concentrations of specified compounds)
selenium	10 µg/l
tetrachlorethene	40 µg/l
trichlorethene	70 µg/l
trihalomethanes	100 µg/l (total) for chloroform, bromoform, dibromochloromethane & bromodichloromethane (after 10 years)(150 µg/l within 5-10 years).
vinyl chloride	0.5 µg/l

(Note: the parameters and parametric values listed above are taken from COM(94)612

and may be amended before the new Directive is finally approved)

Table 10 - Some of the main changes to parameters and parametric values listed in the proposed new EC Drinking Water Directive

Parameter	Change
acrylamide	new parameter with limit of 0.25 µg/l
antimony	reduced from 10 µg/l to 3 µg/l
arsenic	reduced from 50 µg/l to 10 µg/l
benzene	new parameter with limit of 1 µg/l
boron	guide level value replaced with limit of 1 mg/l
bromate	new parameter with limit of 10 µg/l
cadmium	no change
chromium	no change
copper	reduced from 3 mg/l to 2 mg/l
cyanide	no change
1,2-dichloroethane	new parameter with limit of 3 µg/l
fluoride	limit of 1.5 mg/l unchanged, but now irrespective of temperature
lead	reduced from 50 µg/l initially to 25 µg/l then to 10 µg/l
mercury	no change
nickel	reduced from 50 µg/l to 20 µg/l
nitrate	no change
nitrite	no change
PAHs	composite parametric value of 0.2 µg/l retained, but specific limit for benzo(a)pyrene set at 0.01 µg/l
selenium	no change
tetrachlorethene	new parameter with limit of 40 µg/l
trichlorethene	new parameter with limit of 70 µg/l
trihalomethanes	new parameter with limit of 100 µg/l (total) for chloroform, bromoform, dibromochloromethane & bromodichloromethane (after 10 years)(150 µg/l within 5-10 years).
vinyl chloride	new parameter with limit of 0.5 µg/l

3.2 Previous VAM studies

During recent years, a significant amount of information relating to requirements for reference materials has been generated by LGC, both through specific studies undertaken as part of the 1994-1997 VAM Programme and through personal contacts and enquiries received through LGC's Office of Reference Materials (ORM) and the Reference Materials Advisory Service (REMAS). The VAM 1994-1997 study to assess priority reference material requirements [11] involved an extensive survey of industry, government and academia, and culminated in the production of a detailed prioritised list of new and challenging certified reference materials to be produced under

VAM. A composite list of environmental (land and soil) reference materials identified during the VAM 1994-1997 prioritisation exercise and from enquiries received by the ORM/REMAS is summarised in Table 11.

Table 11 - Environmental (land and soil) reference materials identified during the VAM 1994-1997 prioritisation exercise and from enquiries received by the ORM/REMAS

MATRIX MATERIALS	
Matrix	Analyte(s)
Water (generic)	low level nutrients e.g. nitrate, nitrite, phosphate, silica
Water (generic)	sodium, potassium, ammonium, magnesium, calcium, fluoride, chloride, bromide, nitrite, nitrate, phosphate
Water (generic)	dissolved organic carbon, dissolved organic nitrogen
Water (generic)	total nitrogen, total phosphorus
Water (generic)	PAH's
Water (generic)	PCB's
Water (generic)	phenols
Water (generic)	Red List pesticides and less common pesticides
Water (generic)	VOC's
Water (generic)	heavy metals (ppb levels)
Water (generic)	metals
Water (generic)	inorganic anions
Water (generic)	organics
Water (generic)	organic parameters - (a) Red List Compounds & (b) volatile organics
Water (generic)	chromium, copper, nickel, lead, zinc, mercury
Water (generic)	cyanide, phenols, ammoniacal nitrogen
Water (generic)	mercury compounds (speciation) at ppb level
Water (generic)	selenium compounds (speciation)
Water (generic)	TOC
Water (generic)	oil
Water (generic)	colour scan
Water (generic)	carbon 20-50ppm, nitrogen 10ppm, hydrogen 1ppm
Water (generic)	trace, major and heavy metals
Water (generic)	Na ~ 130 mg/l, Ca ~ 120 mg/l, K ~ 16 mg/l, Mg ~ 30 mg/l, HCO ₃ ~ 150 mg/l, Cl ~ 150 mg/l, SO ₄ ~ 390 mg/l
Water (generic)	suspended solids (total as well as dissolved solids)
Water (generic)	dioxins and dibenzofurans
Water (drinking)	PAH's

MATRIX MATERIALS	
Matrix	Analyte(s)
Water (fresh)	plutonium (0.05-0.5 Bq/kg), ²³⁷ neptunium (0.1-1 Bq/kg), ⁹⁹ technetium (0.1-1 Bq/kg)
Water (natural)	trace elements
Water (potable)	microorganisms (e.g. E. Coli, E. Cloacae, Pseudomonas aeruginosa, Coliforms, Cryptosporidium, Campylobacter, Legionella, giardia, cell culture)
Water (potable)	range of typical potable water samples
Water (potable)	hard and soft for metals particularly manganese
Water (potable)	metals, nutrients, PAH's, trihalomethanes
Water (river)	heavy metals
Water (river)	metals, nutrients, (range of types)
Water (sea)	metals, nutrients, (range of types)
Water (sea)	low level nutrients e.g. ammonium, phosphate, nitrate, nitrite, silicon dioxide
Water (sea)	nutrients
Water (sea)	chloride
Water (boiler)	heavy metals
Water (effluent)	high COD effluent for metals, PAH's, PCB's, pesticides
Water (effluent)	high COD & BOD for suspended solids
Water (effluent)	metals and oils
Water (effluent)	industrial
Water (effluent)	domestic
Water (leachate)	inorganic parameters
Water (leachate)	organic parameters
Water (sewage effluent)	metals, nutrients
Water (waste)	organics
Water (waste)	effluent standards, COD, suspended solids
Water (waste)	Red list compounds
Water (waste)	trace elements
Water (waste)	metals, COD, BOD, solids, nitrogen, phosphorus, sulphate, fluoride
Water (waste)	COD
Water (waste)	charring tests, COD, suspended solids
Contaminated Land	free and total cyanide, PAH's, PCB's, OC's
Contaminated Land	elemental sulphur, total sulphate, petroleum hydrocarbon
Contaminated Land	phenols (coal tar and chlorinated phenols)
Contaminated Land	sulphide
Contaminated Land	cyanide and heavy metals

MATRIX MATERIALS	
Matrix	Analyte(s)
Contaminated Land	PAH's and mineral oils
Contaminated Land	heavy metal leachates (matrix)
Contaminated Land	complex cyanide and free cyanide, elemental sulphur
Contaminated Land	phenols, VOC's
Soil	inorganic anions, metals, nutrients
Soil	complex cyanides
Soil	PAH's, PCB's, phenols, VOC's, pesticide residues
Soil	total arsenic and arsenic ^{III} , total chromium and chromium ^{VI} , total nickel
Soil	total and free cyanide, total phenols, free sulphur, total sulphate, total sulphide, total fluoride
Soil	semi-volatile organic compounds
Soil	for all common contaminants and toxic elements
Soil	ammonium, nitrate, phosphate, potassium, total nitrogen, phosphorus
Soil	gallium, arsenic, selenium, fluoride, sulphate
Soil	toluene extractables (set of soils)
Soil	for sulphur (3 types sandy, loamy, clay-peaty) at low, medium, high level
Soil	for physical analysis
Soil	total petroleum hydrocarbon
Soil	DDT, Dieldrin, Lindane, Hexachlorobenzene, Gamma HCH
Soil	sulphate, total sulphate
Soil	hydrocarbons
Soil	Cs, Pd, Pt, Rh
Soil	OP pesticides and PCBs
Soil	organics and loss on ignition(2-4%)
Soil	metals certified using ISO aqua regia extraction
Soil	Trichlorethylene, 1,1,3, Trichloroethane
Soil/silts	²³⁷ neptunium (0.1-1.0 Bq/kg), ⁹⁹ technetium (0.1-1 Bq/kg)
Soil/sediment	organotin compounds
Soil/sediment	pentachlorophenol
Soil/sediment	triazine compounds
Soil/sediment	metals, ammonia, nitrate, nitrite, silicon, phosphorus, sulphate, chloride
Sediment	OC, pyrethroid pesticides, organophosphorus compounds
Sediment	heavy metals, selenium speciation
Sediment	trace elements at low levels e.g. cadmium
Sediment	HCH, DDT, DDE, DDD

MATRIX MATERIALS	
Matrix	Analyte(s)
Sediment	Aroclor - 1016
Sediment	Aroclor - 1221
Sediment	Aroclor - 1232
Sediment	Aroclor - 1242
Sediment	Aroclor - 1248
Sediment	Aroclor - 1254
Sediment	Aroclor - 1260
Sediment	heavy metals
Sediment/sludge	DDE, DDT, TDE(DDD), Dieldrin
Sediment (freshwater)	OC insecticides
Sediment (freshwater)	PCB congeners, PAHs
Sediment (estuarine)	major and trace elements
Sediment (estuarine)	organic carbon and nitrogen
Sediment (marine)	persistent organochlorine pesticides
Sediment (marine)	PAH's, aliphatic hydrocarbons
Sediment (marine)	organic carbon/nitrogen
Sediment (marine)	PCB congeners
Sludge	metals, phosphate, nitrogen
Sludge	OC pesticides and PCBs
Sewage sludge	linear alkylbenzenesulfonates
Sewage sludge	dimethylammonium chloride
Sewage sludge	sulphur up to 25%
Sewage sludge	dioxins
PURE MATERIALS	
Material requested	Certified for:
range of analytical reference standards for pesticides	purity
cis and trans permethrin isomers	purity
certified standard solutions for OC's and PCB's	purity
range of pure (EPA and other important) PCB's	purity
range of polycyclic aromatic hydrocarbons	purity
organo-tin reference standards	purity

3.3 Materials already available or under development

Clearly, during any assessment of future reference material requirements it is also important to review identified and potential needs against those materials which are already available and materials which are presently under development.

The Certified Reference Materials Catalogue [12] provides a comprehensive listing of materials which are currently available through LGC. Although it is not the intention to reproduce detailed listings from the Catalogue in this report, it should be recognised that this source provides a useful summary of the range of reference materials currently available for environmental analysis. Environmental reference materials (land and water) completed by LGC in the 1994-1997 VAM Programme are listed in Table 13.

In addition, two further reports [13,14] prepared under the auspices of CITAC and ISO-REMCO provide further information on reference materials which are under development. A summary of environmental (land and water) reference materials currently under development is provided in Table 12. Table 14 lists environmental reference materials (land and water) currently under development at LGC as part of the 1997-2000 VAM Programme.

Information on materials currently available, along with details of those currently under development, can also be found on specific producers' (e.g. NIST, BAM) web-sites.

Table 12 - Environmental (land and water) reference materials currently under development (from ISO-REMCO status reports)

Matrix	Analyte(s)	Producer
sediment	extractable trace elements (sequential extraction)	CEC SMT (BCR)
sediment	methyl-Hg	CEC SMT (BCR)
sediment	butyl- and phenyltin	CEC SMT (BCR)
sediment	mineral oil	BAM(Germany)
sediment (coastal)	trace elements	IRMM (CEC)
sediment (river)	PCB	CEC SMT (BCR)
sediment (river)	PAH	CEC SMT (BCR)
sewage amended soil and terra rossa soil	extractable trace elements (single extraction)	CEC SMT (BCR)
sewage sludge	Cr	CEC SMT (BCR)
sewage sludge	PAH	BAM(Germany)
sewage sludge	PCB	BAM(Germany)
sludges (X2)	trace elements	CEC SMT (BCR)
soil	trace elements	CEC SMT (BCR)
soil	lead (from paint)	NIST(US)
soil	fuel (diesel)	BAM(Germany)
soil	PAH	BAM(Germany)
soil	PAH	NMI (Netherlands)
soil (contaminated)	TPH	BAM(Germany)

Matrix	Analyte(s)	Producer
soil (contaminated)	PAH	BAM(Germany)
soil (highly contaminated)	As, Cd, Cr, Cu, Hg, Ni, Pb and Zn in aqua regia extracts according to ISO 11466	BAM(Germany)
soil (industrial contaminated)	PAH, pentachlorophenol	CEC SMT (BCR)
soil (industrial contaminated)	dioxins, furans, CP, CB	CEC SMT (BCR)
soils	chemical composition	GUM(Poland)
water	organic substances (RI)	GUM(Poland)
water (drinking)	polar pesticides	CEC SMT (BCR)
water (freshwater - lyophilised)	Cr (III) and Cr (VI)	CEC SMT (BCR)
water (freshwater)	Se (IV) and Se (VI)	CEC SMT (BCR)
water (groundwater)	major compounds	CEC SMT (BCR)
water (groundwater)	trace elements	CEC SMT (BCR)
water (seawater)	Hg	CEC SMT (BCR)
water (simulated rainwater)	trimethyllead	CEC SMT (BCR)
urban dust	trimethyllead	CEC SMT (BCR)
sorbent for thermal desorption	chlorinated hydrocarbons	CEC SMT (BCR)
2-MP-powder	isocyanates (2-MP-derivatives)	CEC SMT (BCR)
8 solutions + fly ash	17 dioxins and furans	CEC SMT (BCR)
acetonitrile solution	aldehydes (as hydrazones)	CEC SMT (BCR)
activated charcoal	aromatic hydrocarbons	CEC SMT (BCR)
beach sands and processed materials	major, minor and trace constituents	SABS (South Africa)
fly ash	dioxins, furans	CEC SMT (BCR)
pure 2-aminoanthracene	mutagenicity	CEC SMT (BCR)
pure benzo(a)pyrene	mutagenicity	CEC SMT (BCR)
pure chemical	crystalline DNPH derivatives	CEC SMT (BCR)
pure chemical	crystalline isocyanate -2-MP- derivatives	CEC SMT (BCR)

Table 13 - Environmental (land and water) CRMs produced by LGC
under the 1994-1997 VAM Programme

Material	Properties
Hard drinking waters	Metals
Soft drinking waters	Metals

Hard drinking waters	Anions
Soft drinking waters	Anions
Hackney brick works soil	Metals
Coal carbonisation site soil	Metals
River clay sediment	Metals
Landfill leachate	Metals
Landfill leachate	Anions

Table 14 - Environmental (land and water) CRMs currently under development as part of the 1997-2000 VAM Programme

Matrix	Properties
Water: estuarine	Metals - low levels
Water: estuarine	Metals - high levels
Soil	Dioxins and dibenzofurans
Soil	PCB
Soil	Anions
Soil	PAHs
Sewage sludge	Dioxins and dibenzofurans
Sewage sludge	Metals (total and leachable)
Sediment: estuarine	Heavy metals
Sediment: harbour	Organometallics (tin)
Fly ash (old technology)	Dioxins and dibenzofurans
Fly ash (old technology)	Dioxins and dibenzofurans
Pulverised fuel ash	Heavy metals
Waste oil	PCBs

4. Consultation with interested parties

A consultation exercise has been carried out to seek updated views and information on reference material requirements and, in particular, from the users and suppliers of test kits and portable monitoring instruments.

Aims of the consultation exercise included:

- to identify particular analytical problems encountered when making measurements in the field;
- to gain an improved knowledge into what specific materials would be used for and, therefore, the type of material and level of certification required; and
- to find out more detailed information on the specific properties of the materials required (e.g. concentration of analyte).

Discussions were held with both the manufacturers and vendors of field test kits and portable monitoring instruments and the users of such equipment. A list of the organisations consulted during the study is provided in Annex 1.

4.1 Views of the manufacturers of equipment

4.1.1 Sources of analytical problems in the field

There was a varied response as to where and why problems arose, but several suppliers noted particular problems arising from cross-reactivities and interferents which could impact on requirements for reference materials. These included:

- additives in drinking water which affect several kits;
- additives (e.g. metal complexing agents) to boiler and process water which affect kits' performance (e.g. with respect to the determination of Fe) - not strictly an environmental monitoring application;
- the presence of biocides in process water which often interfere with sulphite kits;
- testing waters for chlorine when bromine is present (inability to distinguish between Cl and Br);
- testing river water for nitrate when nitrite is present (nitrate test relies on converting nitrate to nitrite prior to the colourimetric determination);
- lack of selectivity with ion selective electrodes (ISEs);
- impact of water hardness on kits' performance.

Not surprisingly, several suppliers cited 'incorrect use' as a primary cause of analytical error in the field.

4.1.2 Views on reference materials needs and requirements

The views expressed by manufacturers and suppliers of test kits and portable instruments regarding the need and requirements for reference materials were mixed.

Several were not convinced that the provision of reference materials would be of value and for a variety of reasons:

- test kits are already supplied with check samples which contain the target analyte to ensure that they are operating correctly;

- those undertaking measurements normally submit an actual sample of the soil/water for confirmatory laboratory analysis during the course of field testing;
- matrices (particularly soil) are so varied that it would be impossible to replicate accurately all of the types that might be involved, particularly with regard to the range of analytes and interfering species that might be present;
- users were unlikely to purchase reference materials because of their limited shelf lives and the risk of contamination when used in the field;
- those undertaking measurements would be unwilling to pay the additional costs associated with purchasing materials (one supplier noted they had previously supplied reference materials, but that these had been discontinued due to a lack of demand).

A common view expressed by a number of suppliers was that the use of reference materials was unlikely to increase unless there was some legislative driver similar to that in the US where the USEPA stipulated the need to use NIST materials. However, some suppliers felt that more serious users might use reference materials on the basis of a recommendation by LGC. One supplier felt that reference materials themselves should only be used under closely controlled conditions which are rarely, if ever, achieved in the field.

Conversely, several of those contacted were very positive about the need for reference materials and some felt that, although reference materials were not used widely in the field by kit users, they potentially had a major role to play for the kit manufacturers, particularly with regard to validation of test kits and methodology. Some manufacturers also noted that reference materials could be not only used internally, but that suppliers might also sell such materials to users, particularly for validating methods and demonstrating compliance with accreditation and quality system requirements.

One supplier indicated that traceable reference materials are used during product development, during manufacture and testing, and during the course of field use - primarily to calibrate sensors and instruments rather than to validate results. In this case, the materials were purchased from an accredited UK supplier and the feeling was that available materials were adequate.

A range of specific types of materials were suggested and these included:

- aqueous environmental matrices (various) - containing metals;
- industrial effluents - containing metals, phosphate, nitrate ammonia, COD, total N and P etc. (as stipulated in the Urban Wastewater Directive);
- industrial process waters (not strictly an environmental application) - containing phosphate, sulphite, nitrite, molybdate, chloride, organophosphonate, hardness, alkalinity etc.

As far as the required analyte concentrations were concerned, these were varied, but it was felt that for environmental applications a general guideline may be obtained from the levels stipulated in the legislation governing the respective analytes (e.g. the Urban Wastewater Directive, the Water Quality Regulations etc.). The main requirement was for concentrations at various percentages of the maximum levels and at these maxima.

The issue of interfering species is complex and is known to cause problems with analysis in the field (see above). It was therefore felt that some reference materials should contain specific, known species whilst others should be free from these, but this would depend on the material's intended use.

With regard to prices, it was felt that existing prices varied greatly according to the nature of the reference material and the purpose of the measurement. As a rough guide, however, it was felt that LGC would need to charge similar prices to those of other suppliers.

4.2 Views of the producers of data

4.2.1 Sources of analytical problems in the field

As with the manufacturers and suppliers of test kits and instruments, the response from users of such equipment regarding why and when analytical problems arose was varied.

Several commented that there were very few problems and that good agreement had been obtained between data obtained in the field and that obtained in the laboratory. Conversely, others were able to cite specific examples of where problems had arisen and also noted that there could be considerable differences between results obtained from laboratory analysis and test kit data.

Examples of specific problems that could impact on requirements for reference materials included:

- presence of manganese causing high readings of free chlorine in water;
- reagents for ammonia tests having limited shelf lives;
- problems with analyte extraction, particularly when not water soluble;
- cross reactivities of ISEs, for example, Na^+ and K^+ ions affecting the response of ammonium ISEs;
- apparent errors caused by particularly moist samples and large grain size during the use of kits to determine PAHs in soil;
- variations between instruments from different manufacturers;
- problems when measuring suspended solids, results being found to vary according to the nature of the suspended material.

Although particular problems had been experienced with older colour chemistry test kits (e.g. for the determination of petroleum hydrocarbons and PCBs in soil) due to cross-reactivities and interferents, it was felt that the newer more selective immunoassay based kits generated data that was fit for purpose.

The incidence of 'false positives' could cause problems when performing analysis in the field, particularly because it was often not possible to distinguish between those false positives which arose from cross-reactivities and interferents (e.g. poor specificity) and those which were due to the malfunction or failure of equipment.

4.2.2 Views on reference materials needs and requirements

Views on the need for reference materials were varied. Although several of those contacted did not perceive a need for reference materials, mainly for the same reasons expressed by the suppliers of instruments and test kits, a number were strongly supportive.

One area where reference materials were highlighted as being required for on-site use was for the analysis of contaminated land, particularly with regard to the use of kits for the determination of PCBs, oils and PAHs, and to a lesser extent explosives (e.g. RDX and TNT) and dioxins. Required concentration ranges were quoted as follows:

- PCBs @ 2-10 ppm
- Oils @ 100-1000 ppm
- PAHs @ 50 ppm

The reason for field testing of contaminated land is to ascertain whether the concentration of contaminants is above or below specific action limits and, in turn, to make a decision on the level of remediation required. As the extent of remediation required not only has environmental, health and safety ramifications, but also cost implications, reference materials which enable more reliable

measurements to be made around decision levels (e.g. to determine whether levels of PAHs are < or > 50 ppm and < or > 1000 ppm) could have significant value. It was felt that reference materials comprising various soils containing varying concentrations of the analytes in question would be of benefit both to validate kits' performance in the field and also during operator training.

Portable instruments, for example, hand-held multi-parameter monitors, are used widely to determine a set of six standard water quality indicators: pH, DO, conductivity, turbidity, temperature and ammonium. Some felt that reference materials were not required, as portable instruments are checked with calibration solutions (e.g. pH buffers) in the laboratory prior to deployment in the field. However, others felt that because environmental conditions, particularly temperature, could affect measurements, and that some form of field calibration was desirable. One example mentioned was turbidity measuring instruments which are checked in the field with a gel containing a particulate suspension prior to each measurement.

5. Analysis and discussion of findings

It is clear that many of the analyses performed using portable measuring equipment are not undertaken exclusively in the field. Samples are often also submitted to the laboratory for analysis, either using similar equipment to that employed in the field or for confirmatory tests using conventional instrumentation and techniques. The development of reference materials which are aimed exclusively at out-of-laboratory applications is therefore unlikely to be viable nor the most effective utilisation of resources and it is considered that materials which have both in- and out-of-laboratory uses will have both the widest applicability and benefit.

5.1 Regulatory drivers

From the sample of documentation reviewed, it is readily apparent that existing legislation and regulation along with national standards and guidelines rely heavily on the ability to make valid and comparable measurements. The relevance and impact of regulation as a driver for reference materials has been illustrated and reinforced during discussions with users who themselves have identified a need for specific materials which are representative of analyses carried out in support of legislation (e.g. the Urban Waste Water Treatment Regulations and the Water Supply Regulations) and national standards and guidelines (e.g. ICRCL limits).

There is also evidence to suggest that future regulation and legislation will significantly impact on reference material requirements, not only in respect of changes to the range and specified limits of parameters covered, but also in relation to the way in which measurements are carried out. This is illustrated in the proposed changes to the EC Drinking Water Directive which, rather than requiring specific methods of analysis to be used, permits the use of any method which satisfies certain pre-defined performance criteria. This change will make it easier for those making measurements to select methods and equipment that are suited to their needs, and will also encourage research into new technology and improved methods of analysis - something which is evidently taking place and illustrated by the rapidly increasing availability of new test kits for environmental monitoring. However, the change also has implications for the traceability of measurements and comparability of results. As comparability is achieved through traceability to either a reference method or a reference material (e.g. CRM), any move which decreases the requirement to use reference methods will place a corresponding greater emphasis on reference materials as the means of ensuring comparability of results. Reference materials which are representative of real samples will also be required in order to validate methods and ensure that specified performance criteria (accuracy, precision and limits of detection) are being met.

Changes to monitoring procedures can also impact on the types and properties of reference materials required. In the new Drinking Water Directive, the two types of monitoring proposed (check and audit monitoring) will not only be carried out at different frequencies, but will also involve the analysis of different groups of parameters. Whilst audit monitoring will involve the determination of all parameters listed in the directive, check monitoring will involve the analysis of a smaller range of parameters, albeit more frequently. Although the suite of reference materials required for audit monitoring would, ideally, need to be representative of the full range of parameters listed in the directive, those materials required for check monitoring would only need to cover a smaller range of parameters (e.g. aluminium, ammonium, colour, conductivity, hydrogen ion concentration, iron, nitrate, nitrite, odour, taste and turbidity).

5.2 Analytical problems in the field

The majority of those contacted felt that problems did arise when undertaking measurements in the field, with some reporting significant discrepancies between data generated in the field and data generated in the laboratory using traditional analytical techniques and instrumentation.

Although there were mixed views on the causes of problems, many related to a lack of full field performance validation, particularly in regard to the poor specificity and selectivity of equipment and cross-reactivity to non-target species.

‘False positives’ are viewed as causing a particular problem when performing analysis in the field and it is not always possible to distinguish between those false positives which arise from cross-reactivities and interferents (e.g. poor specificity) and those which are due to malfunction or failure of equipment. A ‘blank’ material, certified as being free or containing less than a specified concentration of particular analytes could prove useful in identifying and eliminating ‘false positives’. Such a material might also provide a means of identifying sources of contamination during field analysis and also during sampling and submitting samples for laboratory analysis.

Whilst some users recognised that reference materials could have an important role in identifying and overcoming many of the problems experienced in the field, it is clear that much still needs to be done to increase awareness amongst many suppliers and users of equipment of the benefits and role that reference materials have in achieving reliable and comparable measurements.

5.3 Usage of reference materials

The current usage of reference materials for out-of-laboratory measurements made using portable measuring equipment is generally poor, with very few materials being used routinely during field testing, either with kits or instruments.

There is no evidence of colourimetric or immunoassay kit users employing reference materials in the field, except for the standards supplied with the latter which facilitate calibration. However, some suppliers noted that several more technically aware users of colourimetric kits purchase reference materials to validate the instruments used with the kits, but this is not a particularly frequent or routine practice and is not usually conducted in the field. Some kit suppliers noted that, although reference materials have been offered for sale, they have attracted very little interest from the users.

Both the users and manufacturers confirmed that very few reference materials are used with portable instruments, with the exception of pH buffers and turbidity standards, which are used to calibrate the instruments in the laboratory or field prior to deployment. Some of the major instrument manufacturers claimed to use considerable quantities of reference materials, but these generally appeared to be calibration standards or reagents (e.g. water containing known amounts of ammonia, nitrate, chloride, trace metals etc.) rather than ‘real’ reference materials (matrix or otherwise) produced through a formal certification process.

Views on the need to use reference materials for out-of-laboratory measurements made using portable measuring equipment were varied. Several of the suppliers and users of test kits and portable instruments did not consider reference materials to be of value, whereas others clearly recognised their benefits and had strong views on specific requirements.

Several test kit users felt that reference materials would be beneficial and could be used to validate extraction procedures and eliminate/identify any matrix-specific effects. Others felt that using the standards supplied with the kits, combined with laboratory validation on a proportion of samples, was adequate.

A number of suppliers thought that additional materials would be very useful and saw a need for the common environmental analytes in several matrix types, for example, aqueous effluents, process and environmental waters. Conversely, others felt that, as with soil testing, the differing types of matrix would make the preparation of a useful range of reference materials all but impossible.

Whilst there is a plethora of European and UK legislation dependent on reliable and comparable measurements, there does not appear to be a regulatory requirement to use reference materials for field-based measurements using test kits or portable monitoring instruments. Although formal accreditation [15,16] requires the use of appropriate reference materials and calibration standards,

there do not seem [17] to be many organisations which are accredited for chemical testing/analysis of land and water in the field. The absence of a regulatory driver (compared with the USEPA's requirement to use NIST materials) is viewed as a key reason for poor uptake and use of reference materials during field-based testing. The cost of materials, however low, combined with the additional time and effort involved in their use, also detracts from routine field use.

5.4 Types and properties of reference materials required

5.4.1 Generic considerations

Whilst 'out-of-laboratory' reference materials need to exhibit similar properties to those used for laboratory-based measurements, materials intended for 'out-of-laboratory' applications might also require additional properties, particularly with regard to their stability and suitability for use under the range of environmental conditions encountered in the field.

Environmental matrices (soil, sediment, sludge) are particularly prone to degradation (e.g. decomposition etc.) when stored in their 'natural' state and this makes the preparation of stable and homogenous reference materials which are representative of real sample matrices difficult to achieve. One option that could potentially overcome this problem would be to produce dry materials which could be used either in their dry state or 'reconstituted' prior to use (i.e. by the 'addition' of an appropriate reagent). Although this approach might have certain theoretical advantages, for example, it might be possible to vary the composition of the 'addition' reagent and generate 'reconstituted' materials of different composition from a single base material, there are a number of practicalities that warrant further investigation (e.g. the repeatability and reproducibility of reconstitution and mixing).

The issue of interfering species is complex and is known to cause problems with analysis in the field. A suite of reference materials, some containing specific known species, with others free from these, could prove useful. However, several users felt that because solid matrices (soil, sludge, sediment) varied so greatly, both in terms of their matrix and their composition, this would make the preparation of a useful range of reference materials an impossible task. There is a general issue here regarding the perceived usefulness of reference materials, with a number of suppliers and users of equipment appearing to be of the view that, in order for reference materials to be of any value, they need to be identical in composition to the sample being analysed. This is not the case and, although it is highly unlikely that reference materials will ever be identical to actual samples, they have a real value in providing an assurance that methods and equipment are functioning correctly, and that results obtained at one location are consistent with those obtained previously or elsewhere. Clearly, more needs to be done to increase awareness amongst suppliers and users of test kits and portable instruments of the benefits of reference materials, particularly with regard to ensuring the accuracy and precision of methods and equipment used in field.

It is also likely that different types of reference material would be required for different applications. For example, whilst it is likely that more highly characterised certified reference materials (CRMs) would be required for validation and calibration activities, it is questionable whether such materials are required for 'everyday' purposes such as routine analytical quality control. As cost is seen as an important barrier to use, the production of low-cost reference materials (RMs), as opposed to CRMs, should provide an important step forward in facilitating wider uptake.

5.4.2 Specific considerations

Although there was a mixed response as to the general value and need for reference materials for out-of-laboratory measurement, a good proportion of those consulted expressed firm views on specific requirements.

Not unexpectedly, there is a clear link between the views of the types and properties of reference materials needed and the range of analyses required in support of guidelines and regulation. Indeed, when asked about required or ideal analyte concentrations, several of those consulted felt that, for environmental applications, a general guideline may be obtained from the levels stipulated in the legislation.

One area where reference materials were highlighted as being required for on-site use was for the analysis of contaminated land, particularly with regard to the use of test kits for the determination of organic contaminants (PAHs, PCBs, dioxins, explosives - RDX and TNT) in soil at ICRL limits. Other areas where reference materials were seen as being important included the determination of pollutants and contaminants in aqueous samples, particularly, drinking water, river water and ground/surface water.

Specific requirements for new reference materials need to be assessed against both materials which are already available and those which are currently under development. For example, although the study has identified a range of potential materials required for water monitoring (e.g. in support of the proposed new EC Drinking Water Directive), it needs to be borne in mind that a variety of water CRMs (mainly inorganic and elemental analytes) already exist. There are also several water reference materials in the pipeline. Whilst additional reference materials are likely to be required for 'new' analytes, those materials which already exist or are under development may continue to be appropriate for 'existing' analytes, even where there are small changes to the stipulated levels and the concentrations at which analytes need to be measured. Whilst it is clear that there is a need for reference materials containing organic parameters in environmental matrices, the production of stable materials, for example, waters containing trace organics, is particularly difficult.

5.5 Summary of reference material requirements

The earlier sections of this report identify an extensive and diverse range of potential reference material requirements, both in relation to common types of analyses undertaken in support of regulation, and from previous and current consultations on specific needs. It is not the intention to repeat the detailed listings here, but rather to summarise the main groups and properties of reference material requirements identified.

Examples of typical matrices for which reference materials are required include:

- **water:** potable/drinking water, ground/untreated water, river water, estuarine water, sea water, effluent/wastewater and leachates; and
- **land:** agricultural soil, contaminated land, sludge (including sewage), and river, estuarine and marine sediments.

Chemical analytes for which reference materials are required include:

- inorganics: anions, total and trace elements, nutrients, heavy metals and radionuclides;
- organometallics; and
- organics: pesticides, Red List compounds, PAHs, PCBs, VOCs, dioxins and dibenzofurans, phenols, and petroleum hydrocarbons.

Some specific potential reference material requirements identified during the study include:

- wastewater/effluent - BOD, COD, total nitrogen, total phosphorus (concentrations as specified in the Urban Waste Water Regulations 1993)(see Table 5);
- surface (river and coastal) waters - dangerous substances (arsenic, atrazine and simazine, azinphos-methyl, dichlorvos, endosulphan, fenitrothion, malathion, trifluralin, tributyltin, triphenyltin & derivatives) specified in the Surface Waters Regulations 1997)(see Table 7);

- potable/drinking water - new organic parameters (acrylamide, benzene, 1,2-dichloroethane, PAHs - specifically benzo(a)pyrene, tetrachlorethene, trichlorethene, trihalomethanes, vinyl chloride) specified in the proposed new EC Drinking Water Directive;
- potable/drinking water - new/modified trace metals (antimony, arsenic, copper, lead, nickel) specified in the proposed new EC Drinking Water Directive;
- potable/drinking water - new/modified chemical/physiochemical 'Indicator' and 'Check Monitoring' parameters (aluminium, ammonium, colour, conductivity, hydrogen ion concentration (pH), iron, nitrate, nitrite, odour, taste and turbidity) specified in the new EC Drinking Water Directive;
- potable/drinking water - the set of six standard commonly analysed water quality indicators (pH, dissolved oxygen, conductivity, turbidity, temperature and ammonium);
- contaminated land - inorganics (arsenic, cadmium, chromium (vi & total), lead, mercury, selenium, copper, nickel, zinc, boron, cyanide, sulphate, sulphide, sulphur) and organics (PAHs, PCBs, dioxins, phenols, petroleum hydrocarbons) at, where available, ICRL threshold and action limits (see Table 2); and
- contaminated land: a 'blank' soil.

6. Conclusions and recommendations

Analyses performed using portable measuring equipment are not carried out exclusively in the field and samples are often, particularly in critical circumstances, submitted for confirmatory laboratory analysis using conventional instrumentation and techniques. The development of reference materials which are dedicated to out-of-laboratory applications is therefore seen as neither viable nor the most effective utilisation of resources and it is considered that materials which have both in- and out-of-laboratory uses will have both the widest applicability and benefit.

The current use of reference materials, particularly matrix materials, for out-of-laboratory environmental monitoring applications employing portable measuring equipment is generally low with little, if any, evidence of such materials being used with test kits or hand-held instruments. However, there appears to be a far greater use of calibration standards, principally aqueous calibration solutions, which are used widely by test kit and instrument manufacturers during product development, testing and calibration. It is therefore considered that this project should focus on the development of matrix materials, rather than calibration standards which appear to be more widely available.

Existing legislation and regulation along with national standards and guidelines rely heavily on the ability to make reliable and comparable measurements. The impact of regulation on the need for reference materials has been reinforced during discussions with users who themselves have specified the need for materials which are representative of analyses undertaken in support of regulatory compliance. Future regulation will also impact on the range and properties of reference materials required for environmental measurements. Changes to legislation will not only affect the range and specified limits of parameters monitored, but will also impact on the way in which monitoring is carried out. The increasing trend towards permitting the use of any method which meets certain pre-defined performance criteria (e.g. accuracy, precision, limit of detection), as opposed to stipulating the use of specific methods, will augment the role of reference materials in validating procedures and demonstrating the fitness for purpose of the methods employed. It is, therefore, vital that LGC keeps abreast of new legislation and its impact on the requirement for new reference materials, particularly in relation to the continued applicability of existing materials and those currently under development. Where appropriate, and particularly for those reference materials required to underpin European legislation, the possibility of obtaining EU funding should be further explored.

Previous studies and surveys along with personal contacts and enquiries made to the ORM and REMAS have identified a diverse range of reference materials required to support environmental monitoring activities. This study has provided updated information on specific reference material requirements and some of the most popular matrices and analytes identified have been summarised in Section 5.5 above. More detailed information relating to specific requirements is also provided earlier in this report. Whilst the specific materials to be developed under the VAM out-of-laboratory project need to be determined in conjunction with the main reference material prioritisation exercise of project 2.3.1, this study has highlighted two specific reference material activities which warrant further investigation: the production of a certified 'blank' soil material for use in contaminated land analysis; and a practical evaluation of the viability of producing 'reconstituted' soil reference materials.

Even though many out-of-laboratory environmental measurements are undertaken in support of regulation and compliance monitoring, the current use of reference materials in the field is poor. The absence of a regulatory driver is viewed as a key reason for low uptake and use. Views amongst suppliers and users of portable measuring equipment on the benefits of reference materials are varied. Some do not consider reference materials to be of value, whereas others recognise their benefits and have firm views on specific requirements. There are a number of issues surrounding the perceived application and value of reference materials and more needs to be done to increase awareness amongst the suppliers and users of test kits and hand-held

instruments on the benefits of reference materials and their role in achieving valid analytical measurement in the field.

Summary of recommendations:

1. The VAM out-of-laboratory project should focus on the development of reference materials which have both in- and out-of-laboratory uses, as these will have both the widest applicability, benefit and potential uptake.
2. The VAM out-of-laboratory project should concentrate on the development of matrix materials, rather than calibration standards which appear to be more widely used and available.
3. Although the specific materials to be developed under the VAM out-of-laboratory project should be determined in conjunction with the main prioritisation exercise, this study has identified two types of reference material activity which should be considered for taking forward under the project:
 - the production of a certified 'blank' soil reference material for use during the analysis of contaminated land; and
 - the investigation and evaluation of the feasibility of producing reconstituted soil reference materials.
4. There is a general requirement for LGC to keep abreast of regulatory and legislative developments which impact on environmental measurements and, therefore, on the future requirement for new reference materials. There is a specific requirement to determine how the new EC Drinking Water Directive will impact on the requirement for water matrix materials, particularly in relation to the continued applicability of both existing materials and those currently under development. The possibility of obtaining EU funding for reference materials required in support of European legislation should be further explored.
5. There is a general requirement to improve awareness amongst suppliers and users of portable measuring equipment on the application and role of reference materials in achieving valid analytical measurement in the field. At the same time, mechanisms by which regulatory and legislative drivers can be used to stimulate increased demand and use of reference materials need to be explored. These issues should be pursued in other VAM projects, for example, as part of the marketing initiatives being taken forward under projects 1.2.1 and 1.2.2.

7. References

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Annex 1 List of organisations consulted during the study

<u>Organisation</u>	<u>Contact name(s)</u>
AEA Technology (Culham)	Malcolm Caine
Anglian Water	Bob Mathews
Aspinwall & Co	Les Thompson, Nigel Budgen
Cambridge Sensors (formerly Environmental Sensors)	Dr Peter Arnold
Celsis Lumac	Dr Peter Grant
Danes and Moore	Kim Bradley
EI Group/GLI International	Duncan Ward
Environment Agency	Andrew Chappell, Mat Loewenthal
Environment Agency (Midlands Region)	Derrick Smith
Environment Agency (North-East Region)	John Hancox
Environment Agency (South-West Region)	Paul Mitchell
Environment Agency (Thames Region)	John Goddard, Guy Elliott
Guildhay	Romney Jackson
Hach/Camlab	Colin Day
Hydro Data Services	Don Jones
Palintest	Dr Valerie Argent
Rhone Poulenc	Bill Lankford
SDI (formerly Ensys)	Alex Buchan, Colin Green
SWG (Sensors for Water Interest Group)	Mike Scott
Thames Water	Helen Stainer
Tintometer	Dr Mike Pointer, Geoff Shute
W A Fairhurst & Partners	Dr Irene Anders
Yorkshire Water	Pat Bradley, Chris Whitworth